

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 36

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Tropical intertidal coastal forests of halophytic (salt-tolerant) trees in saline, oxygen-deficient (anaerobic) mud (1) • Prop / aerial roots for support in soft sediment and for gas exchange (pneumatophores); salt excretion (1) • Seeds germinate on the parent tree and float, rooting where they touch sand - rapid natural regeneration (1)	3
01.2	<i>1 mark for each point.</i> • Coastal erosion / storm protection: roots and vegetation absorb wave energy (least tsunami damage in 2004 where mangroves were intact) (1) • Fisheries: nursery grounds where dense roots protect young fish that later live on reefs / open sea (1) • Protection of coral reefs: slow river water so suspended solids settle and nutrients are absorbed before reaching reefs (1) • Resources: timber, fuel; medicines (antimicrobials against MRSA, cyanobacterium for lung cancer); high biodiversity (crabs, shrimps, sponges, birds) (1)	4
01.3	<i>1 mark for each point.</i> • Clearance for aquaculture (shrimp ponds), urban development, ports, tourism; pollution; sea level rise / climate change; destruction of coral reefs that shelter the coast (1) • Legal protection / designated areas (eg Sri Lanka protected all remaining mangroves in 2015) (1) • Replanting and natural regeneration; control of damaging activities; Ramsar sites (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 66-67; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> - Natural processes are driven by renewable, low energy-density solar energy captured by photosynthesis; human processes usually use high energy-density non-renewable fossil fuels (1) - Natural processes occur at low temperatures because enzymes lower the activation energy of reactions; human processes need high temperatures and pressures (1) - Natural processes produce no toxic wastes / their products are raw materials for other processes; industrial processes emit CO₂ and NO_x (1) - So natural fixation is sustainable indefinitely with a low carbon footprint (1)	4
02.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Hydrological cycle (evaporation, precipitation) (1) - Carbon cycle / photosynthesis (1) - Atmospheric circulation / winds (1) - Thermohaline circulation (density differences from heating / evaporation) (1) - Decomposition (1)	3
02.3	<i>1 mark for each point.</i> - Bioleaching and phytomining extract metals at low temperature instead of high-temperature smelting (1) - Anaerobic digestion and composting treat wastes; enzymes in industry (biological washing powders, biofuels from algae / microorganisms) (1) - Heat pumps and solar thermal capture low-density heat; low-temperature manufacturing using biological polymers; fuel cells instead of combustion (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
03.1	<i>1 mark for each point.</i> - Coordinating global data on biodiversity conservation (Red List) (1) - Increasing understanding of the importance of biodiversity (1) - Deploying nature-based solutions to global challenges in climate, food and sustainable development (1)	3
03.2	<i>1 mark for each point.</i> - Decline = $(12\,000 - 2100) / 12\,000 \times 100 = 82.5\%$ (1) - Decline over 80% meets the Critically Endangered criterion even though habitat area (400 km²) and adults (2100) only meet Endangered (1) - A species is placed in the highest category for which it meets any one criterion, so Critically Endangered (1)	3
03.3	<i>1 mark for each point.</i> - Categorising a species requires a great deal of research (population, range, trends) and resources; about 85 000 of millions of species assessed (1) - Key species in habitats under particular threat (eg lemurs indicating the state of Madagascar's forests) (1) - Evolutionarily unique species / EDGE species; endemic species; keystone species; flagship species; species with fragmented populations (1)	3

Question	Marking guidance	Marks
03.4	1 mark. Iberian lynx: Critically Endangered to Endangered after captive breeding and protection of two habitat areas (population trebled in 15 years); Fregate Island beetle after rat eradication (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-27; AQA spec 3.1.2.3.1.	
04.1	1 mark for the correct answer. $1000 / 100\,000 \times 100 = 1\%$ (1)	1
04.2	1 mark for each point. - Some light is reflected by the leaves (1) - Some light is transmitted (passes through the leaves without being absorbed) or is of wavelengths not absorbed (1) - Much of the energy captured is used in the plant's metabolic processes and released as low energy-density heat (respiration) (1)	3
04.3	1 mark for each point. - Less energy is passed to each trophic level than was received from the previous one (heat losses in respiration) (1) - After four levels too little energy remains to support a further population (1)	2
04.4	1 mark for each point. - Permanent grassland is a more sustainable land use than arable crops in these areas (soil / rainfall unsuitable for crops) (1) - Grazing animals can digest grass cellulose using symbiotic bacteria in their stomachs, producing food that humans can eat (1)	2
04.5	1 mark for each point. - Pigs are omnivores that can eat food wastes not wanted by people (eg cheese-making wastes, catering leftovers) (1) - So edible meat is produced from material that would otherwise be wasted, without using land / crops that could feed people (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
05.1	1 mark for the correct answer. $2.7 - 2.45 = 0.25$ billion = 250 million years (1)	1
05.2	1 mark for each point. - For millions of years all the oxygen produced reacted with (dissolved) iron in the oceans (1) - Only when all the iron had reacted did surplus dissolved oxygen build up in the oceans and then escape into the atmosphere (1)	2
05.3	Any 3 from the points below - 1 mark each, maximum 3. - Oxygen in the atmosphere absorbed ultraviolet light (1) - A dynamic equilibrium of reactions between O, O₂ and O₃ (the ozone layer) developed (1) - Less UV reached the surface so organisms were no longer restricted to water (which had absorbed the UV) (1) - Aerobic respiration became possible so more complex / larger organisms (plants and animals) evolved; many anaerobic Archaea and bacteria died out (1)	3

Question	Marking guidance	Marks
05.4	<i>1 mark for each point.</i> • Photoautotrophs absorbed carbon dioxide, a greenhouse gas, and some of the carbon was stored in geological sediments such as carbonate rocks and fossil fuels (carbon sequestration) (1) • The lower CO₂ concentration reduced the retention of heat, offsetting the increase in the brightness of the Sun (about 10% per billion years) (1)	2
05.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Biogeochemical cycles developed so small amounts of nutrient elements could support life for long periods without depletion (1) • Transpiration by land plants returned water vapour to the atmosphere and increased rainfall in other areas (1) • Regulation of atmospheric composition (eg oxygen / CO₂ by photosynthesis and respiration) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-8; AQA spec 3.1.1.2.	
06.1	<i>1 mark for each point.</i> • About 40-50 traps (1) • Because the running mean stops changing / fluctuating with fewer than this the mean is unreliable (1)	2
06.2	<i>1 mark for each point.</i> • Widely scattered results (large standard deviation) mean a single sample may not be representative (1) • More samples are needed so that the effect of variability is averaged out and the mean is representative / significance can be assessed (1)	2
06.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Same type and size of pitfall trap, set for the same length of time (1) • Traps emptied at the same time of day / same weather conditions in both habitats (1) • Same number of traps and same method of positioning (random or systematic) in each habitat (1) • Other variables that could affect results (eg rainfall, temperature) monitored or controlled (1)	3
06.4	<i>1 mark for each point.</i> • Flying insect activity changes with weather over minutes to weeks (temperature, wind) and diurnally (1) • Samples on different occasions are needed to produce a mean that represents the typical value or to identify trends (1) • The interval between samples should match the rate at which the factor changes, found from a preliminary study (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
07.1	<i>1 mark for each point.</i> • Its reading is adjusted to match a meter known to be accurate so that the results are accurate / comparable (1) • If it cannot be adjusted the percentage error should be recorded so the results can be corrected (1)	2

Question	Marking guidance	Marks
07.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Droppings: diet, gender, territory size (1) • Owl pellets: diet (1) • Feeding marks (eg chewed hazel nuts): presence of species such as dormice (1) • Nests / burrows: presence and breeding (1) • Tracks / footprints: presence, movements (1) • Territorial marks (scratching posts): presence and territory (1)	3
07.3	<i>1 mark for each point.</i> • Bat detectors / sonograms record the ultrasonic sounds bats produce, which differ between species (1) • The recordings provide evidence of presence, abundance and activity of species that cannot easily be seen (1)	2
07.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Marked individuals mix freely with the rest of the population (1) • Marks do not fall off or wear away, and marking does not affect survival (1) • The population does not change during the study (no births, deaths or migration) (1) • All individuals have an equal chance of being caught / animals are mobile and can be caught (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402, 405, 407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
08.1	<i>1 mark for each point.</i> • Very small gene pool so individuals are closely related and inbreeding is likely (1) • Harmful recessive genes are expressed when both parents carry them: some goslings had thin hair-like feathers with poor insulation so survival in cold weather was low (1) • A stud book records family trees so breeding is arranged between the least related individuals (often coordinated by a specialist zoo) (1) • Exchange of individuals / semen between zoos; avoid hybridisation with related species / subspecies (1)	4
08.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Breeding is triggered by stimuli (day length, light, temperature, food, body fat) which may be unknown / not provided (1) • Population interactions: mixing breeding and non-breeding individuals causes conflict; some need a choice of partners, others pair for life (1) • Breeding habitat needed: courtship display sites, nest sites, social groups (flamingos only breed in large groups - mirrors used) (1) • Stress / disturbance (1)	3
08.3	<i>1 mark for each point.</i> • AI: semen collected (and can be stored / cryopreserved for years) and inserted into the female, so animals need not be transported or meet; avoids dangerous mating (elephants, rhinos) and partner rejection (1) • Embryo transfer: a female of the endangered species (eg bongo) is treated with hormones to release many eggs, which are fertilised and implanted into surrogate females of a common related species (eland) (1) • So many offspring are produced per cycle instead of one calf per year, overcoming the shortage of breeding females (1)	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 33-35; AQA spec 3.1.2.3.3.	
09.1	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Hedgerows (1) • Hay meadows (1) • Grazed moorland / heathland (1) • Chalk grassland (1)	2
09.2	<i>1 mark for each point.</i> • Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive (1) • Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers (1)	2
09.3	<i>1 mark.</i> • The introduced (non-indigenous) predator may eat other species as well as the pest (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: wild catches have not increased since about 1990 as stocks are fully or over-exploited; aquaculture output growing rapidly and approaching the wild catch; fish are the last large-scale hunted food • Advantages of aquaculture: control of limiting factors, selective breeding, gender control, high yields per area; fish have low FCRs (no thermoregulation / support) compared with livestock; herbivores (tilapia, carp) and molluscs very efficient; polyculture, IMTA, aquaponics, rice-fish • Limits: carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins); some species (bluefin tuna) will not breed in captivity; consumer preference for carnivores • Environmental impacts: habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control; and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits; MSY management needs good data • Judgement: aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing; both need sustainable management	9

Question	Marking guidance	Marks
	Model answer - Level 3 (9 marks)	
	Wild catches have not increased since about 1990 because most stocks are fully or over-exploited, while aquaculture has grown rapidly and now approaches the wild catch, so the question is how far it can take over. Its advantages are real: limiting factors, breeding, gender and disease can be controlled, yields per area are high, and fish have low food conversion ratios because they need not support their weight or keep warm; herbivores such as tilapia and carp and filter-feeding molluscs need no fishmeal at all.	
	Its limits are equally real. The carnivorous species consumers prefer - salmon, shrimp - are fed fishmeal from wild forage fish, so their farming still depends on fishing and depletes species such as sandeels on which puffins rely; some species, such as bluefin tuna, will not breed in captivity. Farms destroy mangroves, release organic waste, pesticides and antibiotics that breed resistance, spread lice, allow escapes that contaminate wild gene pools and provoke predator control, though siting, closed systems, polyculture and vaccines can reduce these. Meanwhile fishing itself can be sustainable if quotas based on good stock data, gear design, size and effort limits and no-take zones are enforced.	
	Aquaculture can therefore supply most of the growth in demand, especially through herbivorous fish and shellfish, and can relieve pressure on wild stocks, but it cannot fully replace fishing while carnivorous species depend on wild fishmeal and some species cannot be farmed. Both must be managed sustainably; neither is a substitute for the other.	
	Weaker answer - Level 1 (2 marks)	
	Aquaculture is farming fish instead of catching them. Fish farms can produce a lot of fish and they do not run out like wild fish do. But fish farms cause pollution and diseases can spread to wild fish. Salmon farms feed the salmon on fish which have to be caught. So aquaculture can replace some fishing but not all of it.	
	What makes the difference	
	• Level 1 reaches a reasonable conclusion but supports it with two sentences; Level 3 gives the context (flat wild catch since 1990), the advantages in detail and the impacts with ways to reduce them.	
	• Level 3 distinguishes between species types (herbivores and molluscs vs carnivores) - the key to 'to what extent'.	
	• Level 3 also considers whether fishing can be made sustainable, which the comparison requires.	
10.2	<i>1 mark.</i> • A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	

Question	Marking guidance	Marks
11.1	Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Energy flow: autotrophs and heterotrophs; losses by reflection, transmission, respiration; 100 000 to 1 000 units; few food chains beyond four levels; plant food most productive; grazing animals where crops impossible; pigs on wastes • Energetics: intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • Control of genetics: asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • GM: Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape; advantages; risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity) • Sustainability links: energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which agriculture can be made sustainable while feeding a growing human population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Need: population possibly 12 billion; humans raise carrying capacity; trophic level of diet • Impacts to be reduced: habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • Social, economic and political influences: consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • Strategies: cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • Limits: little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25